

6 (long)

A simply supported timber beam subject to a uniformly distributed load is shown in Fig. 6. Attached to the beam, mid-span, is a vertical cable that passes over a frictionless pulley to support a suspended weight P . The designed weight of P is 300 kN such that the mid-span deflection of the beam is zero when the beam is subject to a uniformly distributed load of 30 kNm^{-1} . The beam has a rectangular cross-section of 600 mm depth and 200 mm breadth. The timber has a Young's modulus of 11 GPa and sustains a maximum longitudinal stress in bending of 24 MPa.

- (a) Draw the shear and bending moment diagrams for the beam subject to this loading arrangement, noting all salient values. [10]
- (b) It is found that the as-built weight of P is 20% less than originally designed. What is the resulting maximum longitudinal stress at the critical section? Is the section adequate? [10]
- (c) What is the mid-span deflection associated with this reduction in the weight of P ? Comment on the magnitude of this deflection and any implications for designers in analogous situations. [10]

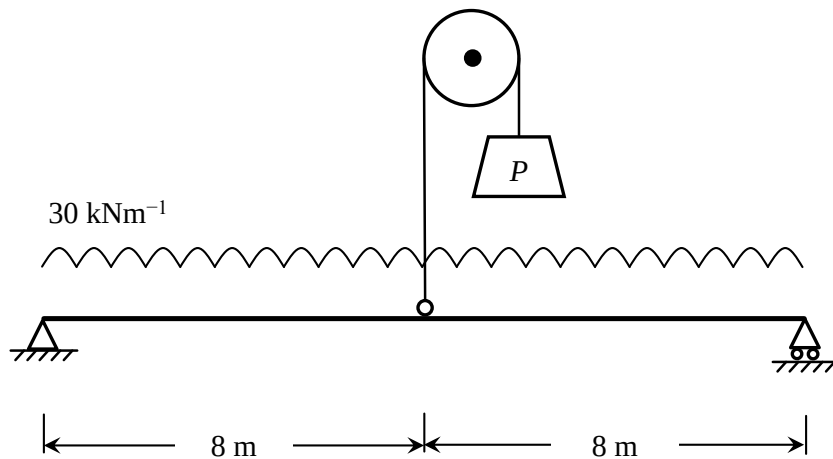


Fig. 6